

Motion2VecSets: 4D Latent Vector Set Diffusion for Non-rigid Shape Reconstruction and Tracking

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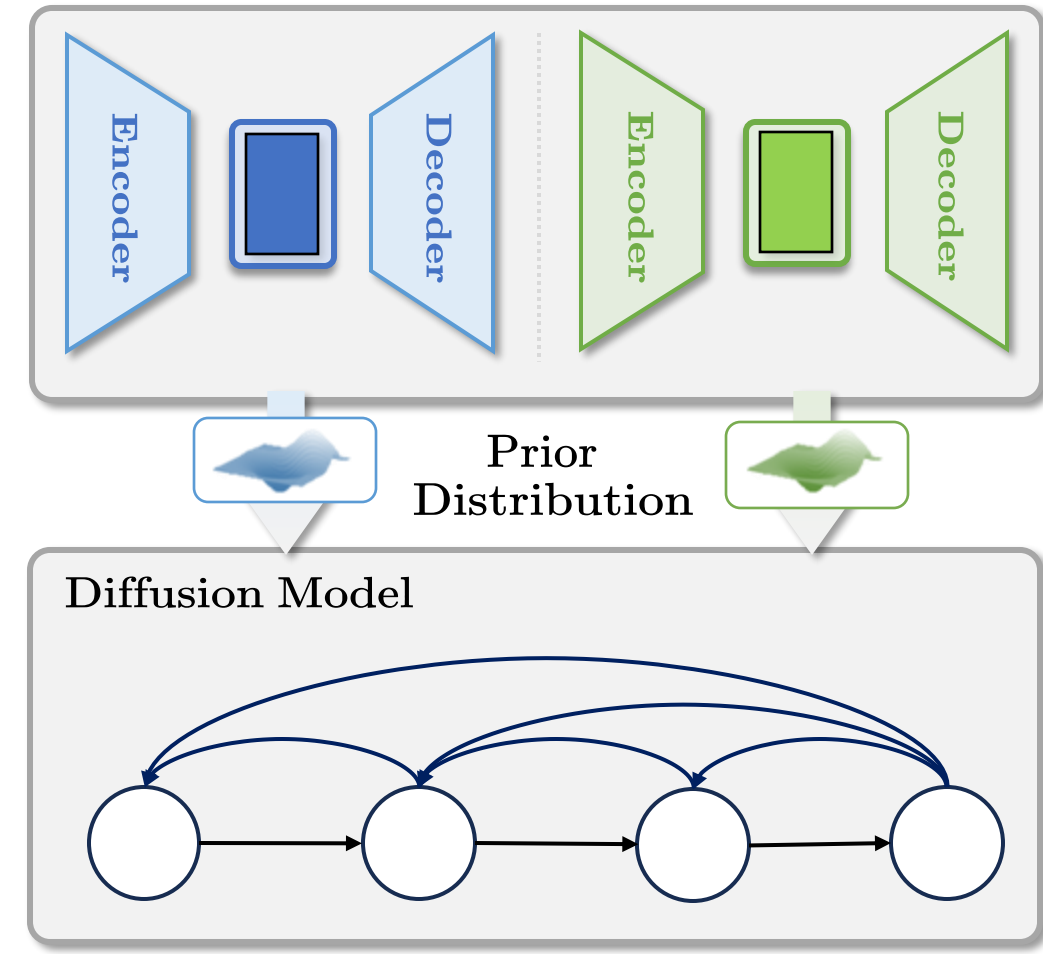
Introduction

◆ Problem

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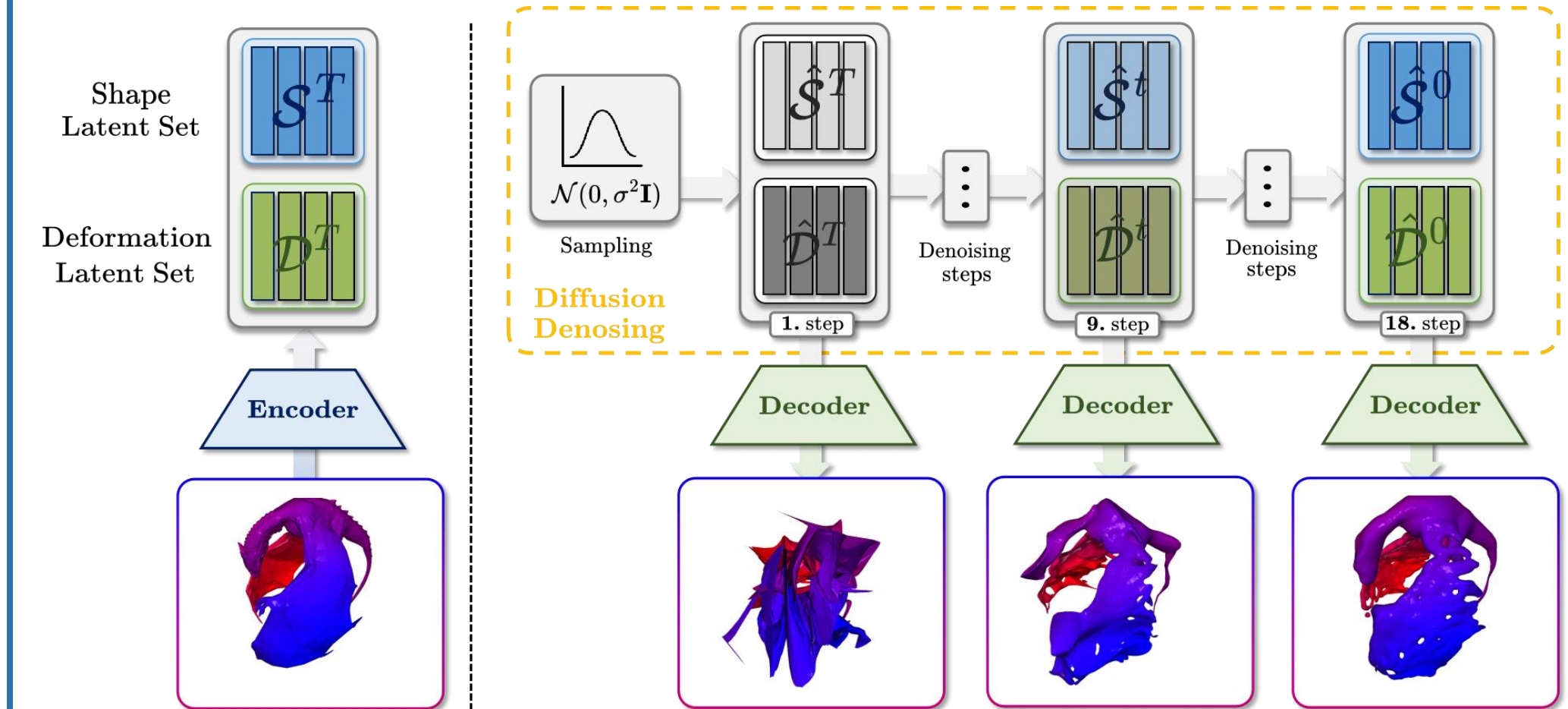
◆ Our Goal?

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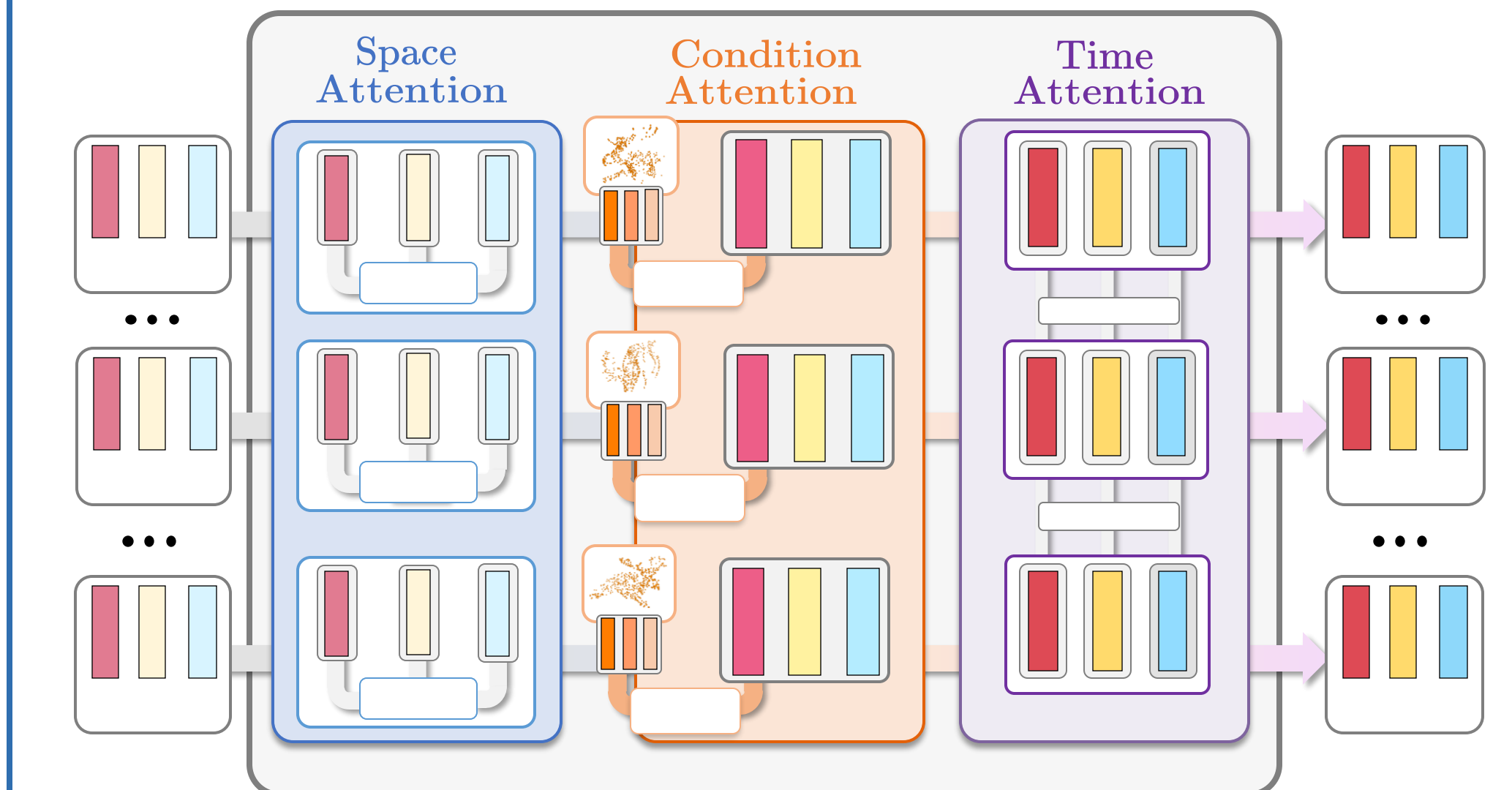


Key Insights

◆ 4D Latent Representation & Diffusion Model

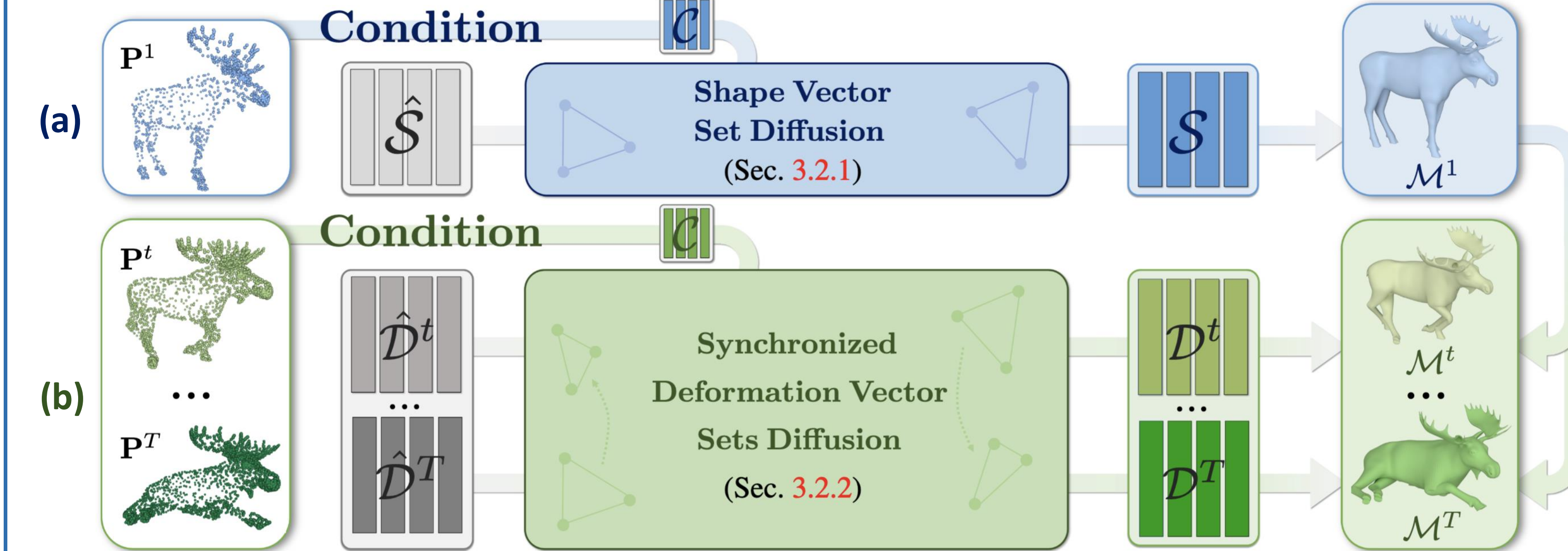


◆ Interleaved Spatio-Temporal Attention Mechanism



Methodology

◆ Overview pipeline of Motion2Vecsets



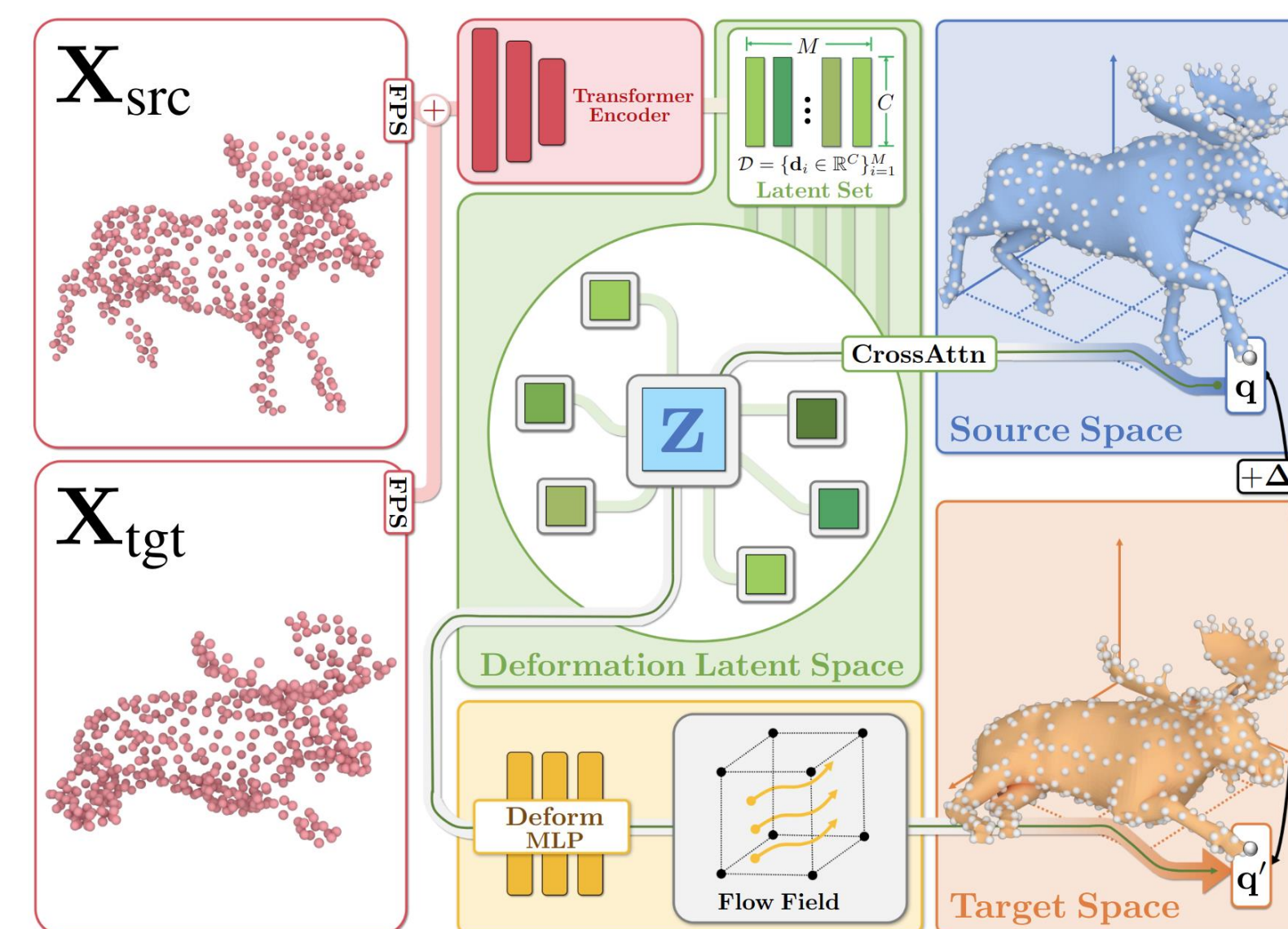
Given a sequence of noisy sparse or partial point clouds as inputs, denoted as $\{\mathbf{P}^t\}_{t=1}^T$. Our model outputs continuous mesh sequence with fine details, denoted as $\{\mathcal{M}^t\}_{t=1}^T$.

(a): The initial input frame $\mathbf{P}^1$ (top left) is used as condition in **Shape Vector Set Diffusion**, yielding denoised shape codes $\mathcal{S}$ for reconstructing the geometry of the reference frame, $\mathcal{M}^1$ (top right).

(b): Concurrently, the subsequent input frames $\{\mathbf{P}^t\}_{t=2}^T$ (bottom left) are utilized in the **Synchronized Deformation Vector Sets Diffusion** to produce denoised deformation codes $\{\mathcal{D}^t\}_{t=2}^T$, where each latent set $\mathcal{D}^t$ encodes the deformation from the reference frame $\mathcal{M}^1$ to each subsequent frame $\mathcal{M}^t$.

◆ Deformation Autoencoder with multiple local deformation fields aggregation

Given a pair of point clouds $\mathbf{X}_{\text{src}}$ and $\mathbf{X}_{\text{tgt}}$ from two frames of a dynamic mesh sequence, we initially downsample them using farthest point sampling. Subsequently, the concatenated points are passed into **transformer encoder** to generate **Deformation Latent Set** $\mathcal{D}$. For a query $\mathbf{q}$ in the source space, a cross-attention layer is utilized to match the most relevant **fused feature** $\mathbf{z}$ is subsequently fed into the **deformation MLP decoder** to predict an offset $\Delta\mathbf{q}$, translating it to $\mathbf{q}'$ in the target space.

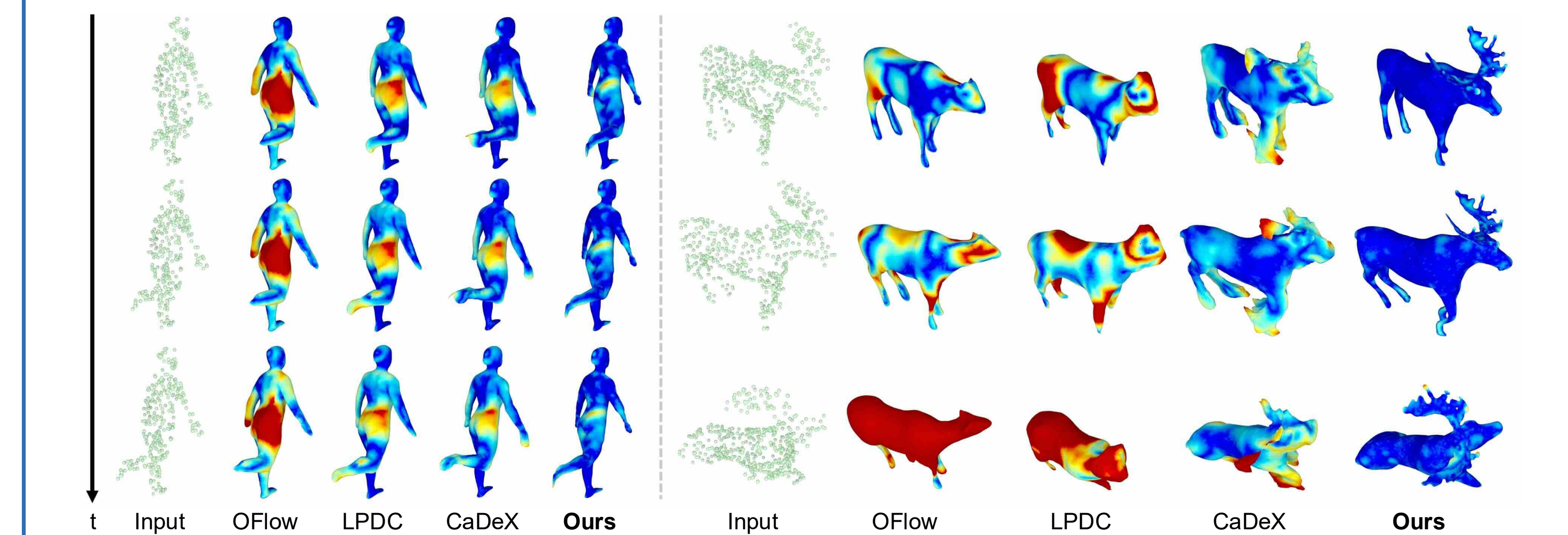


Quantitative Results

Dataset	Method	Sparse Input						Partial Input					
		Unseen Motion			Unseen Individual			Unseen Motion			Unseen Individual		
		IoU↑	CD↓	Corr↓	IoU↑	CD↓	Corr↓	IoU↑	CD↓	Corr↓	IoU↑	CD↓	Corr↓
DT4D-A	OFlow	70.6%	0.104	0.204	57.3%	0.175	0.285	64.2%	0.305	0.423	55.1%	0.408	0.538
	LPDC	72.4%	0.085	0.162	59.4%	0.149	0.262	62.2%	0.339	0.427	51.6%	0.467	0.488
	CaDeX	80.3%	0.061	0.133	64.7%	0.127	0.239	70.8%	0.254	0.499	59.2%	0.379	0.498
	Ours	88.9%	0.050	0.061	83.7%	0.058	0.074	73.3%	0.177	0.404	66.3%	0.193	0.438
DFAUST	OFlow	81.5%	0.065	0.094	72.3%	0.084	0.117	76.9%	0.305	0.423	66.4%	0.109	0.194
	LPDC	84.9%	0.055	0.080	76.2%	0.071	0.098	68.3%	0.339	0.427	59.6%	0.156	0.204
	CaDeX	89.1%	0.039	0.070	80.7%	0.055	0.087	80.7%	0.254	0.499	70.4%	0.096	0.157
	Ours	90.7%	0.033	0.047	83.7%	0.045	0.064	83.8%	0.177	0.404	74.4%	0.075	0.140

Qualitative Results

◆ Sparse Input Setting



◆ Partial Input Setting

